Maren

CONFIDENTIAL

Maren Finance
Progress & Evidence

What has been built so far, deployed where, proven how, and the exact commands that let you check every claim without trusting us.

DOCUMENT	DATE	VERSION	PREPARED FOR
MRN-100	17 August 2026	1.0	Mezo Ecosystem

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1

What Maren is, in one paragraph

Maren is a consumer borrowing and payments app for Mezo, on web and iOS. Users deposit Bitcoin, borrow MUSD against it from as little as 0.001 BTC, and spend it: pay a `@name`, share a payment link, request money. It is built for Mezo and deployed nowhere else. Everything below is what exists today, with the evidence.

2

Deployed contracts, Mezo matsnet (31611)

Five contracts, all source-verified on Blockscout, deployed from `0x54196397D6f01b48739dB3957257cB33cD341885` in 11 transactions costing 11,688,063 gas in total (about 4 nano-BTC).

CONTRACT	ADDRESS	RUNTIME BYTES
MarenVault	`0x46252B00AB59384518d9dB6D1f210ffF1221645b`	14,508
MarenPay	`0xe475Ff919D54E055809f516E3bed8a5c7a101836`	9,264
MarenRelayHub	`0x22cC1Adfb0C91D9359746dE3F7E8cB7F2B8eA98C`	7,934
MarenRegistry	`0x9e123Cdd0CD524812CF9F552d3FAB13d8143b77E`	4,742
MarenEarn	`0xE3B02b215085B99Ce61E0F5ea9763Ac637aa7826`	11,396

MarenRelayHub extends the ERC-2771 forwarder and is the single trusted forwarder; `MarenPay.trustedForwarder()` returns its address on-chain.

3

Exercised on-chain, not just deployed

- **A pooled MUSD trove is open and active inside Mezo's own TroveManager**, at 0.069 BTC collateral against 2,001.9 MUSD of debt: an ICR of **220%**, with borrowing enabled. Verify:
  `cast call 0xE47c80e8c23f6B4A1aE41c34837a0599D5D16bb0 "getTroveColl(address)(uint256)" 0x46252B00AB59384518d9dB6D1f210ffF12216`
- **A live borrow is offered and correctly bounded.** From the funded test account, a 30 MUSD borrow simulates clean and a 40 MUSD borrow reverts with `InsufficientCollateralRatio` at the account's own 180% floor. The interface offers exactly the range the contract accepts, computed before any signature is requested.
- **The `@sele` handle is registered** through commit-reveal: `resolve("sele")` returns the deployer address on-chain.
- **User flows executed on the live chain:** Bitcoin deposits, a withdrawal, a keeper collateral top-up, and an on-chain payment request of 0.50 MUSD. Each one is a row in the app's Activity screen with its transaction hash linked to the explorer, because the app reads history from contract logs and renders nothing it cannot prove.

4

Test evidence

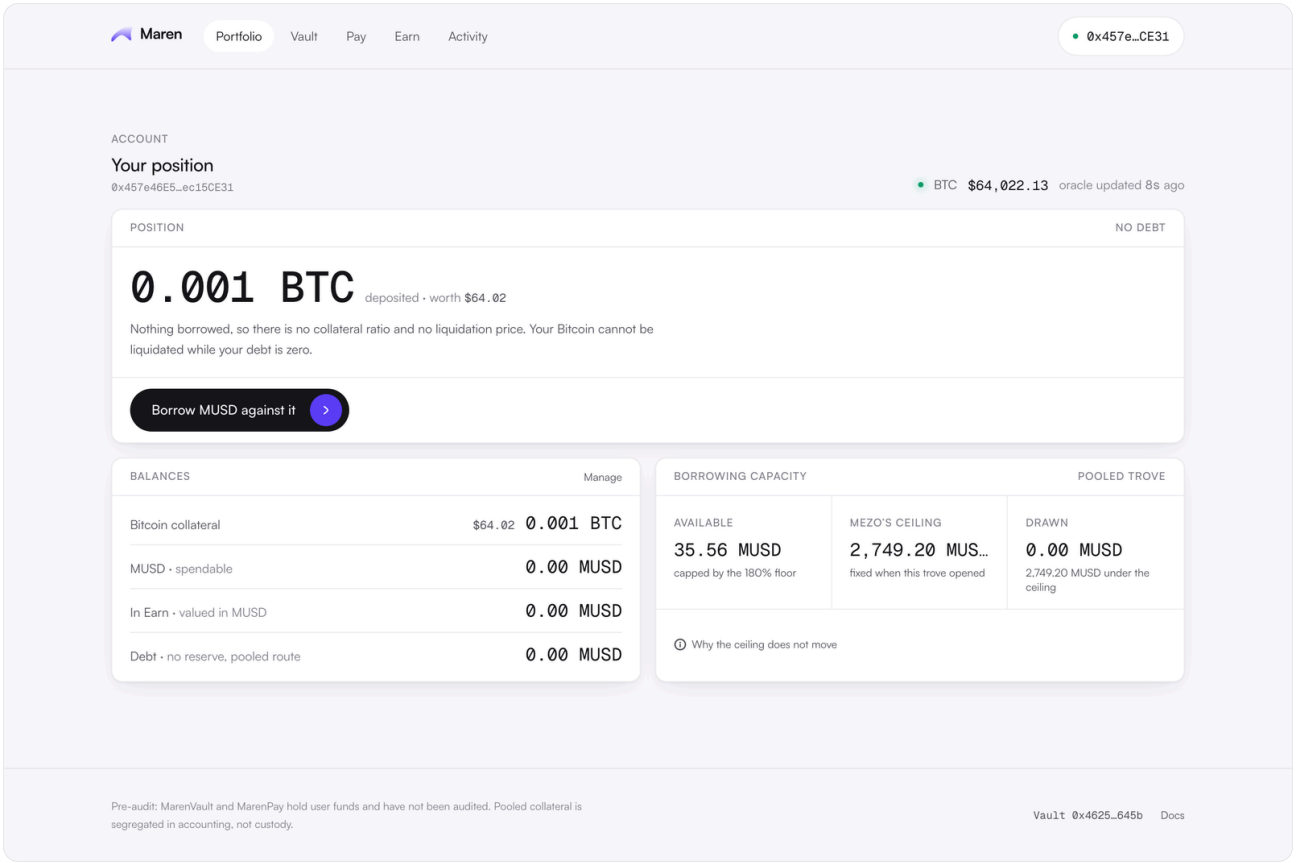
SUITE	COUNT	RUNS AGAINST
Contract unit and property tests	236	Foundry
Fork tests	8	The real MUSD contracts on matsnet
Relayer	214	Four adversarial review rounds, each of which found and fixed a real defect
Web behaviour suite	87	The built app against live matsnet, through an injected wallet that structurally cannot sign

The behaviour suite deserves one sentence of explanation: it drives the real interface against the real chain with a read-only signer, asserts that every offered action matches what the deployed contract would accept, and proves that no signature is ever requested. The account nonce is read before and after as the final check.

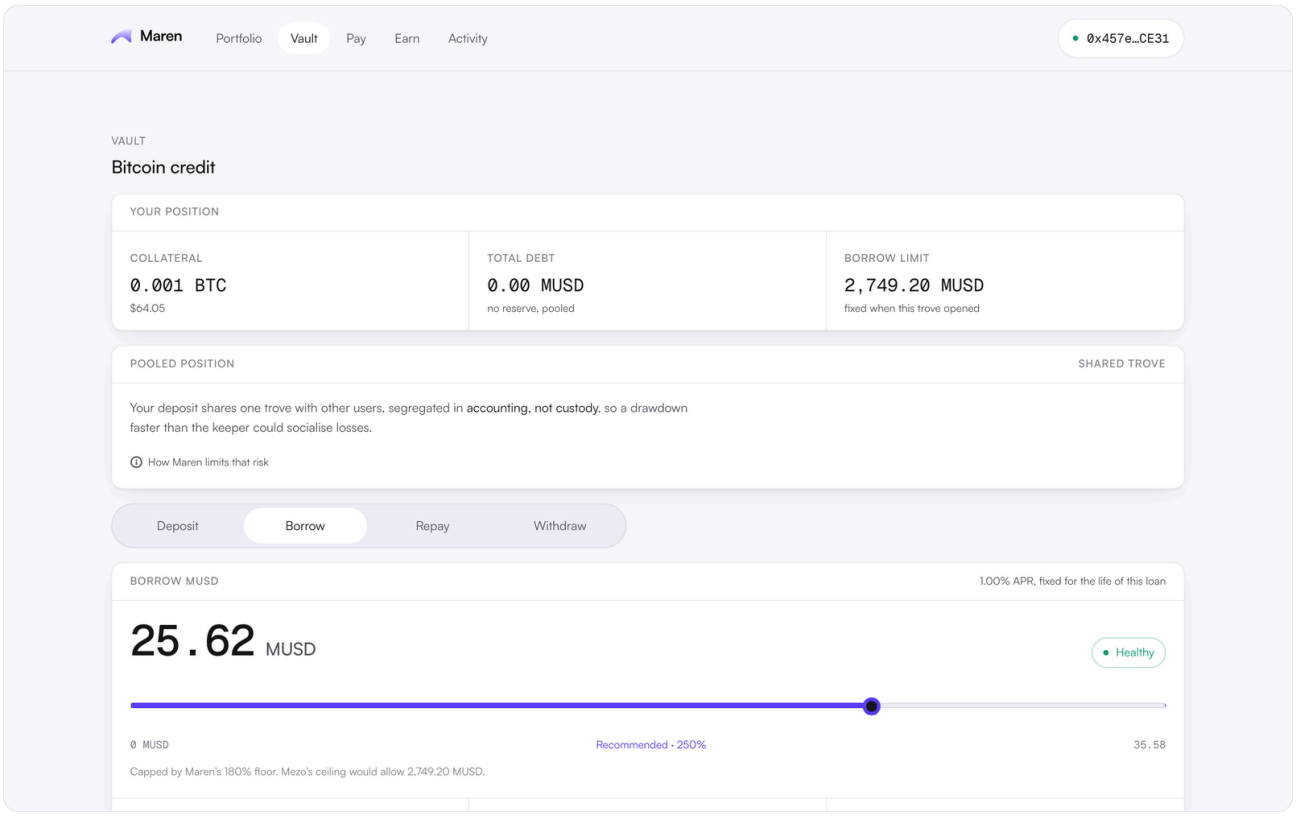
5

The product, as it runs today

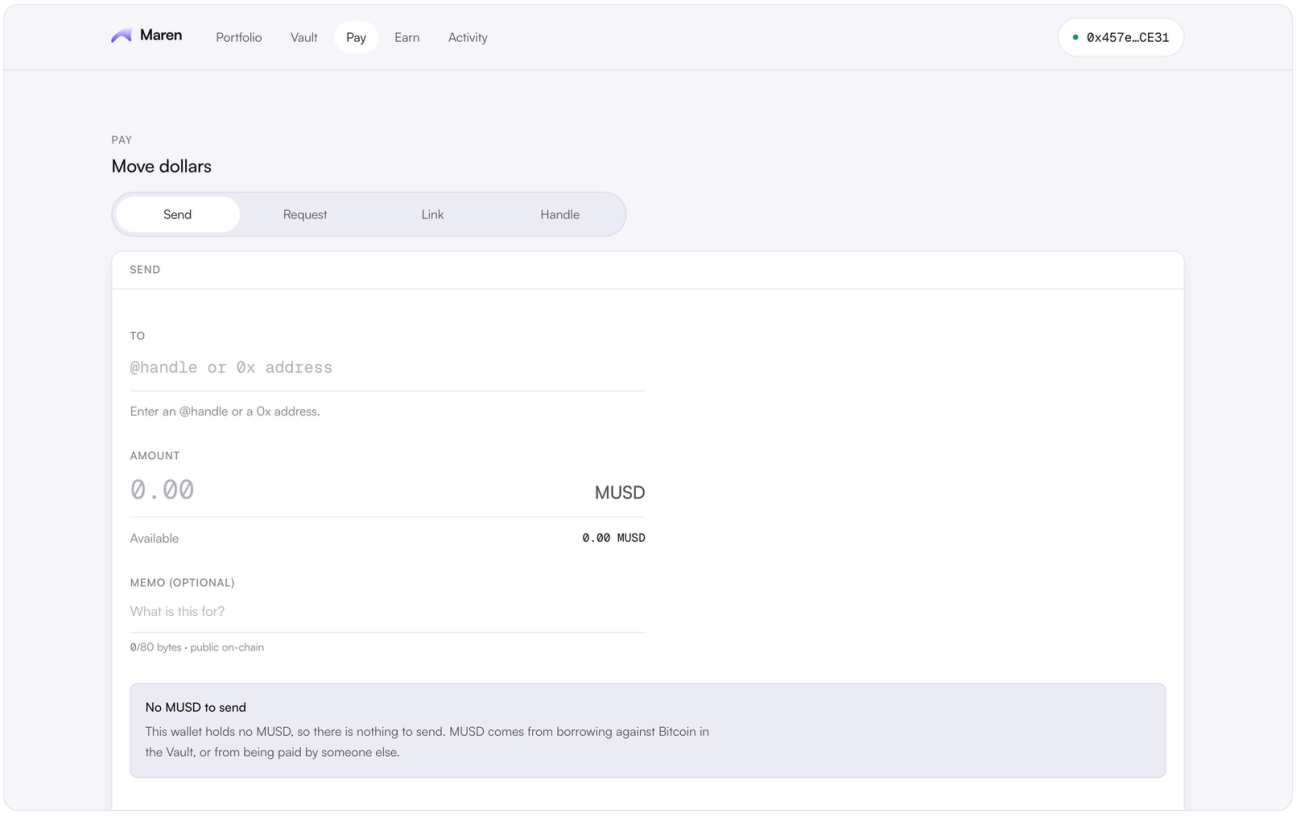
Unretouched captures of the web app in a read-only session against live matsnet, 17 August 2026. Every figure shown is a contract read.



Portfolio. A funded account with no debt: 0.001 BTC deposited, and the interface states plainly that nothing can be liquidated while debt is zero.



Vault. Borrowing live against the pooled trove: the slider is bounded by the same computation the contract enforces, with the recommended 250% marked.



Pay. MUSD moves against @handles or raw addresses, with an on-chain memo. An empty wallet is told where MUSD comes from rather than just refused.

The iOS app is built with Expo, 18 screens, verified in the simulator including the stale-oracle state.

6

Findings from reading the chain

This work surfaced several things that are useful to Mezo regardless of anything else:

1. `viem`**'s Mezo chain definition** ships a default RPC that does not resolve and is missing `multicall3` on mainnet, which silently degrades batched reads for every wagmi user on the chain. A fix is drafted.
2. **The subgraph documentation** says `network: mezo-mainnet`; Goldsky's actual slug is `mezo`, so a manifest copied from the docs is rejected at deploy. Fix drafted.
3. **Three RPC hosts in the developer docs do not resolve.** Fix drafted.
4. **Two places the MUSD docs and deployed bytecode disagree:** the borrower redemption-fee waiver does not appear in `TroveManager`, and the refinance fee computes to roughly 0.02% rather than the documented 0.1%.
5. **ERC-4337 is not viable on Mezo today.** EntryPoint v0.7 is deployed byte-identical to Ethereum L1 but has processed zero UserOperations per the block explorer index, and the RPCs do not implement `debug_traceCall`, so no compliant bundler can run. Maren uses EIP-2612 permits with ERC-2771 relaying instead, and the chain accepts EIP-7702 type-4 transactions as the upgrade path.
6. **A sizing result worth passing on:** a pooled vault seeded at the protocol minimum can never serve a borrow, because MUSD's 1,800 minimum net debt pins the seed below its own borrowing floor. The minimum viable pooled vault is larger than the minimum trove. We hit this in practice and resolved it by sizing the seed to policy.

7

Not done, stated plainly

No mainnet deployment yet: it is gated on an audit, deliberately. No audit yet. The gasless relayer is built and fully tested but not yet hosted, so today users submit their own transactions and the app says so on every screen. Nothing in this document overstates that.

8

Reproduce every number

All chain figures above can be re-read in one pass with `cast` and `curl` against `https://rpc.test.mezo.org` (matsnet) and `https://jsonrpc-mezo.boar.network` (mainnet). The feasibility appendix carries the full command list; the addresses in this document are sufficient to verify independently.

Every chain figure in this document was read live from Mezo RPCs on 17 August 2026, and each one is paired with the command or address that reproduces it. Counts that drift daily are quoted at the precision at which they hold. Where Mezo's published documentation and the deployed bytecode disagree, this document follows the bytecode and says so.